

Aniruddha Deshmukh

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EDUCATION

Vellore Institute of Technology, Vellore

2023 – 2027 (Expected Jul 2027)

B.Tech in Electrical and Electronics Engineering · CGPA: 9.01 / 10

- Coursework: Embedded Systems, Microprocessors & Microcontrollers, Control Systems, DSP, AI/ML, Digital Electronics

EXPERIENCE

Invendis Technologies India Pvt. Ltd., Bangalore

May – Jun 2026

Hardware Design Intern · R&D Division

- Designed the schematic for a Wi-Fi 6 router/gateway (MT7981A Filogic 820) for the Silbo product line in OrCAD Capture 17.2 — 15 organised pages spanning power, Ethernet, Wi-Fi RF, USB, DDR4, and 5G modem subsystems.
- Engineered multi-rail power distribution (4 synchronous bucks, 3 LDOs, RC-delay sequencing) with component selection driven by datasheet calculations, derating analysis, and thermal checks; integrated a 7-port GbE switch, DDR4 termination, Wi-Fi 6 RF frontend, and 5G modem interfaces.

RESEARCH & PUBLICATIONS

Cross-Maze Generalisation in Autonomous Navigation Using NEAT

2025 – 2026

A. Deshmukh, S. Brahmanekar, A. Vinod, M. Subashini — VIT Vellore · IEEE RICE 2026 [Under Review]

- Compared genome aggregation against multi-environment training for NEAT-based drone navigation across 100 procedurally generated mazes; aggregated 100 independently evolved controllers via innovation-number-aligned weight averaging with a 30% majority threshold.
- On an unseen test maze, the aggregated controller survived 26% longer (338 vs 268 frames) and reached 29% closer to the goal (331 px vs 464 px) than the multi-environment baseline; identified three failure modes informing NEAT-GRU extensions.

PROJECTS

PEARL — Physical Environment Aware Reasoning Layer

2026

7th Semester Capstone · Embedded AI Diagnostics · Supervisor: Dr. Jacob Raglend I

- Developed bare-metal STM32F401RE firmware (HAL, PlatformIO) for multi-sensor acquisition: MPU9250 9-axis IMU over I2C (400 kHz) with AK8963 magnetometer bypass, ACS712 current sensor via 12-bit ADC, bidirectional UART at 115200 baud.
- Implemented I2C bus recovery — automatic detection of stuck-SDA conditions and clock-line toggling to restore communication without power cycling.
- Designed a project-agnostic hardware description layer (YAML) that decouples sensor topology from inference logic, enabling hardware swaps with zero code changes.
- End-to-end pipeline: STM32 → UART (JSON telemetry, 5 Hz) → Raspberry Pi 5 → three-layer context engine → LLM (Gemini Flash via API, or Qwen 2.5 locally via llama.cpp) → live FastAPI/WebSocket dashboard.
- Demonstrated real-time diagnosis of four fault scenarios — I2C bus disconnection with auto-recovery, magnetic interference, physical disturbance, and orientation change — distinguishing faults by which sensor channels deviate.

Tech: STM32 HAL (C), Python, FastAPI, WebSocket, PySerial, Gemini API, llama.cpp, YAML, PlatformIO, Raspberry Pi 5

Self-Describing Hot-Swappable Sensor Modules with Universal TinyML Engine

2025 – 2026

Modular Edge ML · ESP32 · EEPROM-Resident Neural Networks

- Trained a 1D CNN gesture classifier (ESP32 + MPU-6050, 99.2% accuracy, 4 classes); built a custom C++ inference engine from scratch (Conv1D, MaxPool, Dense, ReLU, Softmax), eliminating the TFLite Micro dependency.
- Designed a 512-byte binary descriptor encoding NN architecture, sensor configuration, and class labels in module-resident EEPROM; a universal ESP32 runtime parses the architecture and loads float32 weights from a 24LC512 at runtime.
- Python toolchain packs trained models (architecture + weights + metadata) into 64 KB EEPROM images with CRC-32 verification; the end-to-end prototype reads a 7-layer model (57 KB) and classifies live sensor data in real time.

Tech: ESP32, 24LC512 EEPROM, MPU6050, C/C++ (PlatformIO), Python, I2C, TinyML

Neural Network DC Bus Voltage Control for Grid-Tied PV-BES System

2025 – 2026

Special Project · VIT Vellore · MATLAB/Simulink · dSPACE

- Replaced the PI outer voltage loop of a bidirectional DC-DC converter with NARX, Vanilla RNN (with trim integrator), and LSTM controllers trained on PI response data; the LSTM cut steady-state V_{dc} error from 6.98% to 3.02%, RMSE from 5.61 V to 5.05 V, and ripple from 23.76 to 20.14 V_{pp} versus PI on a 150 V DC bus.
- Validated on a laboratory hardware setup (dSPACE real-time controller, PV emulator, LiFePO₄ battery) with an SN-LMS-controlled single-phase inverter; measured grid-voltage and grid-current THD of 1.8% and 3.1% in both grid-following and grid-supporting modes.

Sim-to-Real Robot Controller — Domain Randomisation for RL Transfer

2025 – 2026

Reinforcement Learning · PPO · PyBullet · Domain Randomisation · PID Baseline

- Implemented PID and PPO (stable-baselines3) controllers for an Ackermann-steering robot in a custom PyBullet/Gymnasium environment with fixed-seed benchmarking; across 8 noise conditions, PPO success rate dropped from 88% to 72% under harsh noise while PID held at 68–74%.
- Applied domain randomisation during training, which flattened PPO's performance spread from 20 points (72–92%) to 6 points (72–78%) — matching PID's consistency — but did not improve worst-case success, suggesting the harsh-noise gap is not purely a training-distribution problem.

Real-Time Acoustic Localisation Turret

2026

Distributed Embedded Systems · TDOA · MEMS Microphones · ESP-NOW

- Built a pan-tilt turret that orients toward impulsive sounds: 4 INMP441 MEMS microphones sampled at 44.1 kHz across two ESP32 nodes (ESP-NOW data link + GPIO sync pulse), with adaptive per-mic noise baselines and empirical cross-node gain calibration.
- Estimated pan by blending TDOA cross-correlation with a 41-entry lookup table trained on 2,661 labelled azimuth samples, and tilt from the elevation energy ratio; under 50 ms from event detection to servo command. Demonstrated live at SELECT Makeathon 2026.

PCB Design Portfolio

2025 – 2026

KiCad · Personal Hardware Projects

- Designed and open-sourced three PCBs — a mechanical keyboard, a macropad, and a USB-C Power Delivery hub — with full schematics, layouts, and fabrication-ready files on GitHub.

SKILLS

Programming: C/C++, Python, MATLAB

Embedded: STM32 (HAL, bare-metal), ESP32, Raspberry Pi, Arduino

Design & Simulation: KiCad, OrCAD Capture, MATLAB/Simulink, Fusion 360

ML / Robotics: TensorFlow, TFLite, llama.cpp, neat-python, stable-baselines3, PyBullet, Gymnasium

Protocols: I2C, SPI, UART, I2S, ESP-NOW

TEST SCORES

IELTS Academic: 8.0 overall (Listening 8.5, Reading 8.5, Writing 7.0, Speaking 7.5)

LEADERSHIP & ACHIEVEMENTS

VP of Education & Secretary — Fogo VIT (Toastmasters)

Jan – Dec 2025

- Secretary (Jan–Jun), then VP Education (Jul–Dec) for a 50+ member club; Secretary Trainer in the district Officer's Training Program.
- 3rd Place, Humorous Speech Contest, Coruscate Division Conference (District 120); completed the Dynamic Leadership Pathway (all 5 levels).